Homework 4: Independent Biomedical Data Analysis Project

GSND 5340Q, BMDA – 150 Points Total

Multiple Due Dates (Report + Video: Friday June 19, 2026; Peer Comments: Friday June 26, 2026)

Objective

Take a real biomedical dataset from raw data through interpretation, using one (or more) of the methods we covered this semester. This is a small but **complete data analysis project.** You will present your analysis in a recorded video and provide constructive feedback on your classmates' projects.

Instructions

1. Pick a Dataset and an Analysis Type

Choose one of the following analysis families that we covered in class (or another one you clear with the instructor):

- **GWAS / variant calling** – alignment, variant calling, association or family/segregation analysis, interpretation of disease-relevant variants.
- **Bulk RNA-seq differential expression** – counts, normalization, DESeq2 / edgeR / limma, pathway analysis.
- **Microbiome / metagenomics** – 16S or shotgun reads, taxonomic profiling, alpha / beta diversity, differential abundance.
- **Single-cell RNA-seq** – QC, normalization, clustering, marker/DE.
- **Another -omics method** (proteomics, methylation, ATAC-seq, etc.) – get instructor approval first.

Where to find data:

- A dataset provided in lecture (asthma microbiome, HIV/TB RNA-seq, Ogden's Syndrome variants, TBnanostring, . . .). You may **not** repeat the exact analysis we did in class – pick a new biological question, a new comparison, or a new method.
- A public repository: SRA, GEO, dbGaP, TCGA, GTEx, ENCODE, Qiita, ImmPort, etc.
- A dataset from your own research, with PI permission.

2. R Markdown Report (60 points)

Submit a `.Rmd` (or `.qmd`) that, when knit, walks the reader through your analysis end-to-end. At minimum:

- **Background and question** – one short paragraph describing the dataset, the biological/clinical question, and why it matters.
- **Data acquisition** – how you got the data (download script, SRA accessions, GEO ID, dataset DOI). Include the code or SLURM script.

- **Preprocessing / QC** – normalization, filtering, batch correction, etc. as appropriate.
- **Statistical analysis** – the core method (DESeq2, limma, association test, alpha/beta diversity, clustering, ...) with a brief justification of *why* this method fits your question.
- **Visualization** – at least one publication-quality figure that supports your main finding.
- **Interpretation** – what does your top result mean biologically? Cite at least one supporting paper.
- **Reproducibility** – `sessionInfo()`, links to data, all code included in the Rmd.

3. Video Presentation (40 points)

Record a **5-7 minute video** walking through your analysis. Cover:

- The dataset and your question.
- The pipeline (data → QC → analysis → result).
- Your main finding and one figure that shows it.
- Limitations and reasonable next steps.

Post the video in the Canvas Chat section by **11:59 PM, Friday June 19, 2026**.

4. Peer Review (50 points)

Watch at least **five** of your classmates' videos. For each:

- Name one specific strength of the analysis.
- Suggest one improvement – an alternative method, an additional QC step, a confounder to check, a follow-up comparison.

Document the videos you reviewed and your comments in the same `.Rmd` from part 2. Submit the updated `.Rmd` on Canvas by **11:59 PM, Friday June 26, 2026**.

Grading Breakdown

Category	Points
Reproducible R Markdown report with end-to-end analysis	60
Video presentation with clear walkthrough of question, methods, and result	40
Peer review: meaningful comments on five classmates' videos	50
Total	**150**

Additional Notes

- This is a **data analysis** project, not just a plotting project. Your report must include a statistical or model-based step – not only descriptive plots.
- You may reuse code from lectures or earlier homework, but the question and dataset must be your own, or used in a different way than we did in class.
- Be respectful and constructive in peer reviews – the goal is to learn from each other.

Happy analyzing!